Ayush Agarwal

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Summary

Data Scientist with a strong foundation in advanced analytical research and deep statistical modeling. Authored published, peer-reviewed research on multi-criteria decision frameworks and predictive physicochemical modeling. Specializes in leveraging advanced algorithms (XGBoost, Prophet) and Python/SQL to extract intelligence from complex datasets, solve ambiguous problems, and build ROI-driven financial models.

Skills

Languages & Databases: Python, SQL (Complex Joins, CTEs)
Data Science Libraries: scikit-learn, XGBoost, Prophet, PySpark, pandas, matplotlib
Tools & Cloud: Databricks, Snowflake, AWS, Git/GitHub, Jupyter
Core Competencies: Statistical Modeling, Hypothesis Testing, Time-Series Forecasting, Decision Frameworks

Certifications & Training

- **Snowflake Data Cloud:** Completed foundational technical pathways and earned official data warehousing learner badges.
- **Artificial Intelligence using TensorFlow:** SmartInternz & Google.

Experience

Labcorp Bengaluru, India
Data Analyst Associate Intern Feb 2026 – Aug 2026

- Conducted rigorous exploratory data analysis (EDA) using complex SQL (Window Functions) to isolate statistical anomalies and hidden patterns in real-world datasets.
- Designed and deployed interactive dashboards that translated multi-dimensional data into actionable intelligence, improving the accuracy of internal reporting metrics.
- **Tools Used:** Python, Advanced SQL, Pandas, Scikit-learn, PowerBI

Research & Publications

Predictive Modelling of Physicochemical Properties (QSPR & QSAR) *Letters in Applied NanoBioScience*

- Authored multiple research manuscripts focused on predicting chemical bioactivity using Topological Indices, deploying ensemble regression models (Random Forest, AdaBoost) to optimize computational accuracy.
- **Published QSPR Paper:** [Link to Paper / DOI: 10.33263/LIANBS143.192]

Urban Plant Suitability Index (UPSI) Framework *Environmental and Sustainability Indicators*

- Designed and computed a novel multi-criteria decision-making framework to evaluate tropical urban ecosystems by synthesizing diverse environmental variables into a single statistical scoring metric.
- **Published Paper:** [Link to Paper]

Core Projects

QuickBite Express – High-Performance Optimization Engine *Python, Streamlit, Scikit-learn*

- Engineered an intelligent operational optimization engine utilizing machine learning to streamline delivery dispatching and routing decisions.
- Implemented predictive models to accurately estimate demand surges and dynamic constraints, improving overall processing efficiency.

NYC Rental Arbitrage Engine *XGBoost, SHAP Analysis, Python, SQL*

- Developed an end-to-end predictive modeling engine using XGBoost to identify high-yield investment properties and rental arbitrage opportunities.
- Applied SHAP (Shapley Additive exPlanations) analysis to interpret complex model feature weights, isolating key drivers of market profitability.

SCM-Optima – AI Supply Chain Control Tower *Prophet, Random Forest, Python, Streamlit*

- Built a predictive analytics framework leveraging Prophet time-series models to forecast complex 30-day market demand with high statistical confidence.
- Automated feature engineering pipelines on multi-dimensional datasets to optimize model training inputs and reduce prediction error rates.

Education

Your University Name Bengaluru, India
Bachelor of Technology (B.Tech.) in Your Engineering Branch Aug 2022 – June 2026
Cumulative GPA: **8.69/10.0**